Creating a Web-tool to Produce Site-specific Recommendations for Native Wetland Plant Restoration in the Town of Amherst, NY

By

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I owe this work to my family, friends, professors, colleagues, partner, and cat. I hope that this effort contributes to a better world.

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Abstract

Land use changes and the selection of non-native flora, both largely due to human action, reduce ecosystem resilience and biodiversity. Conservation organizations encourage the planting of native species, but it can be difficult for homeowners to choose plants that match their property conditions. In this project, I develop a tool that uses wetland designations acquired from New York State GIS databases to create a public, interactive, native plant recommendation service. This project is designed for Amherst, New York, a suburban town with about 130,000 residents and 11 mi^2 of wetland. Plant selections are informed by soil type and wetland status, combined with hydrographic data, and then analyzed with multiple ring buffer proximity completed in ArcGIS. A wetland proximity score is derived through summed buffer ranking and linked to native species information from the New York Flora Atlas. By matching geographic information and their plant type preferences, users obtain results for native plants likely to perform well under the indicated conditions. This project explores the concept of connecting the public to ecological restoration information in a succinct and inexpensive way. Reproducibility of the website's structure, educational potential, and public accessibility are major considerations in development to enable widespread use of the strategies outlined in this paper.

Public Abstract

Local environmental conservation through native plant restoration is limited by a lack of public access to specific knowledge. This project is an open-source interactive webpage where users can find suitable native plant types for their property in Amherst, New York. Maps including soil type and wetland proximity are included in an interactive mapping webtool. The webpage is coded through R, and data on plants native to Erie County are filterable by name and several plant traits.

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1. Introduction

This project is the development of a publicly accessible and reproducible website and associated web-tool designed to inform residents of the Town of Amherst of their capacity to grow native wetland plants. Understanding ecosystem resilience and restoration practices in relation to plant suggestions is necessary for more effective results from the web-tool. The host site's function also is to inform the decision-making process behind the tool, which for transparency and reproducibility is developed in R through RStudio. The ability to interact with and educational capacity of the site are major considerations in its development. Full efficacy of the web-tool lies in the access to and suggestion of native plant recommendations it provides to residents.

Ecosystem resilience is the ability of an ecosystem to maintain structure and function through disturbances (Walker 2020). If one organism becomes non-functional, another may take its place and maintain the productivity of the ecosystem without crowding out other functionally similar organisms (Vasiliev 2022). Existing connections to, and spatially distinct locations of, an ecosystem also contribute to the resilience of that ecosystem type, given an extreme event such as a fire where biota needs recovery (Vasiliev 2022). An ecosystem's stability improves when increasing its biodiversity and thereby its resilience to stress (Vasiliev 2022). Loss of ecosystem connectivity through human land-use change damages the range and mobility of various species. Ecosystem restoration is possible and improved by reestablishing native plant species.

A focus on vegetation restoration is important due to the lower capacity of plants to disperse from one place to another. Animals have conscious agency in their relocation, where the seeding or other dispersal methods of plants is constrained by the location and environment of those plants. The plant has little to no conscious agency in their final destination, as gravity, wind, water, or animal dispersal act on the seeds to spread them. Due to these methods of movement, vegetative restoration is hampered by the removal of emerging plant species which are often seen as weeds in otherwise preferred lawn landscapes. In addition, climate change can affect how plant dispersal occurs (Clark et al. 2014). This includes shifts in range of successful plant establishments. Studies of vegetation such as on tree sapling species shifts in powerline corridors show a change to more southerly varieties of plants (Treyger and Nowak 2011). Lack of effective dispersal due to changing climactic conditions and established species' seed dominance also contributes to vegetation's inability to shift effectively (Liang et al. 2018). Vegetation with specific environmental requirements are less likely to find a suitable location, which can affect the connectivity of habitats and result in less robust ecosystems (Churko et al. 2020). Human activities causing fragmentation and habitat loss are more impactful than climate change impacts alone, and thus human intervention in a restorative response is required (Hess et al. 2024).

Human understanding of the local environment is often based on incomplete knowledge, leading to unwise actions. For example, people in the United States often culturally accept short grass cover and decorative non-native trees as attractive and desirable traits (Minor et al. 2016; Burr et al. 2018). This socioeconomic status of the perfect lawn and standardization as a point of standing and acceptance creates tension between residents and disincentivizes change (Burr et al. 2018). Intentional planting and subversion of the monocultural lawn by including multiple species (e.g., clover) and native plants support greater biodiversity among local pollinator populations (Burr et al. 2018). Given the prevalence of lawns as a landscape default after the development of a plot of given land, the distance between native patches of plants grows with every expansion of human-managed area.

While it is difficult to organize and implement native plant restoration on the scale that would solve the problem of fragmentation, even minor changes to local landscapes are preferable (Churko et al. 2020; Spieles 2022). Interested parties who can make the necessary changes are important for the feasibility and potential benefits of restoration to others (Chazdon et al. 2024). Thus, the capacity to know and choose native plants for property owners' lands and homes becomes a very important stage of conserving and restoring local biodiversity. Resources for property owners to effectively reintroduce native plant species can be obscure both in documentation and accessibility (Saran et al. 2022). Appropriately organizing information in an accessible way can jump-start the process for those seeking to combat biodiversity loss (Saran et al. 2022). Nurseries and standard gardening and landscaping plans may stock and recommend popular non-native plants as they are recognizable to customers (Cavender-Bares et al. 2020). Due to there being few alternatives, or less common knowledge of them, native plants are not chosen. Partial restoration of local ecosystems is preferable to no change and is quite possible with the correct information (Churko et al. 2020). Property owners deciding on plant species seem to have no comprehensive and interactive web tool for native restoration based upon their local conditions and needs.

This project puts restoration decisions in people's hands, letting them choose native plants that will best benefit from their property's ecological condition. In restoration, wetland areas are of particular concern as they provide important water filtration and climate mitigation in addition to being bastions of biodiversity (Janse et al. 2019). Due to their incompatibility with common human land development, wetland flora are the aim of this project's recommendations. The utility of this project is in the access it provides, collecting information which would require significant time and effort to otherwise obtain. Adaptability of the webtool as well as free public access to it are therefore considerations in the creation of this project. With appropriate computer data storage, creating a tool which can act as a skeleton for repeated user questions is possible. The project also aims to act as a template that can be copied in any other location. Ideally, just changing base files and plant data will allow user experiences to be similar, lowering barriers to planting native species.

The goal of this project is to develop a searchable map supported tool which effectively allows residents of the Town of Amherst to navigate to their property. Wetlands and wetland proximity values are displayed on the map as well as soil names and hydric type. Finally, using knowledge of their property condition and filtering for desired plant traits, users are able to discover native plant recommendations at their location.

2. Literature:

2.1 Wetland Biodiversity and Connectivity

Given that recent human development has led to an estimated 50% of physical wetland area lost in the 20th century, not enough data exists for confident predictions of how quickly biodiversity might recover on a longer timeframe (Pezzati et al. 2018). Biodiversity recovery in wetlands is estimated to require an incredibly long scale, with passive restoration taking up to one hundred times longer from active restoration (Pezzati et al. 2018). This indicates that the total removal of a wetland environment may not be rectified effectively by passive means.

Land use and land cover change significantly reduces the number of wetlands and reduces hydrological connectivity. Wetlands provide significant extreme event protection against droughts and floods, serving as a buffer and filter for surface water in these events (Janse et al. 2019). Much of the land use and land cover (LULC) change is of human origin and can severely lessen the capacity of wetlands to serve these purposes as well (Man et al. 2024). Increasing the hydrological connectivity of wetlands improves upon methods which only focus on species migration resistance in wetland biodiversity networks (Qu et al. 2024). Introducing a hydrological vegetation gradient can create more efficient water use by placing plants in order of water needs (Qu et al. 2024). This is an indication of healthy wetland connectivity, and in combination with constructed wetland and restored wetland practices greatly increases the water processing and biodiverse capacity respectively of both types (Spieles 2022).

2.2 Public Interaction with Native Biodiversity

Involving the public with the environment around them and reducing the separation manufactured by LULC changes should be an important step of instilling conservation literacy. The official designation of protected wetland areas is associated with increased public awareness of these and other wetlands (Do et al. 2015). Limited interactions with environmental education may not result in marked improvements over time despite short-term changes in perception and action (Champine et al. 2023). People's willingness to change their behavior regarding biodiversity practices also required a lack of barriers to participation (van Heezik et al. 2020). Time commitment, appropriate space, and social pressures for community neatness can all become barriers to actual behavioral change (van Heezik et al. 2020). That does not mean that increases in interaction are impossible, as in the van Heezik et al. study, nearly half of the participants continued with further activities after the initial instruction period. Creating a better understanding of people's relationships with plant life as a community of practice is critical to incorporating environmental values (Beckwith et al. 2022; van Heezik et al. 2020). Other attempts at creating interaction, such as public exposure to botanic gardens, show some promise in marking parallels between human activities and the interactions of species in an ecosystem (Giovanetti et al. 2020). Longer periods of exposure and a narrative of normativity and community assist in making biodiverse practices a habit (Beckwith et al. 2022; van Heezik et al. 2020).

Increasing public knowledge requires an understanding of what the public already knows and thinks about biodiversity and native plants. Residential garden diversity is a necessary starting point of analysis, as understanding how people structure their gardens in the present can inform on their planting behavior in the future. The needs, values, and interests of households are reflected in gardens, which often serve as an extension of the effort and resources homeowners are able to spend tending them (Delahay et al. 2023). Improved connectivity of ecosystems may also contribute to the biodiversity within, yet the individuality of households and lack of consistency in approach make it difficult to reproduce and quantify actual benefits (Delahay et al. 2023). Difficulty also lies in the individual perception of aesthetics, as biodiversity does not conform to ideal and orderly plans (Mikkonen and Raatikainen 2024). This does not preclude all attempts to increase biodiversity in an aesthetic manner, but increases the difficulty in assessing what would be acceptable to residents in terms of planting initiatives.

The shifting climate has some impact on what types of vegetation are considered beneficial to plant, suggesting that a strong narrative in favor of native planting is a critical part in acceptance. Given non-native options with better climate adaptation, there is evidence suggesting that people would choose these plants (Hoyle et al. 2017). Factors such as aesthetics, locational context, inevitability, and perceptions of invasiveness and native incompatibility also determine attitudes towards introduced plants (Hoyle et al. 2017). Emotional responses to vegetation management seem to be more important than logical conclusions about perceived improvements (Straka et al. 2022). Change from what plants are nostalgic and anthropomorphized, from what is culturally accepted and propagated, and in relation to perceived distance from oneself, have been identified as factors in individual feelings about species management (Straka et al. 2022). Complex analysis and balance of these factors, among others, suggest that area-specific management practices will require more individualized expression and alteration to function ideally (Russo et al. 2025).

Existing plant diversity depends in part on the availability of plant species in an area. Commercially available plants in an area can be strikingly representative of local species composition, including those that seed spontaneously (Cavender-Bares et al. 2020). Belief in perceived benefits of various plants was also skewed toward commercially promoted, often exotic plant life (Cavender-Bares et al. 2020). Increasing attention to native plants' benefits in the US has amplified demand for them and their production in recent years (Rihn et al. 2022). Larger varieties of plants are sold by green industries that sell native plants, and the practices they undertake reflect some challenges of establishing sustained profit. This includes longer growth times, lower production count, more stringent pest management needs, best practices of reproduction, and requirements of high quality given those limitations (Rihn et al. 2022).

Sourcing native plants is also of concern, particularly in places as regulated as New York State. Regulations against harvesting any plant materials on state land, as well as against propagation or reselling of state provided native plants from any nursery, and licensing requirements for nurseries and other growers under agriculture and market law are all complications that must be considered by prospective buyers and sellers in the state ("NY State Senate Bill 2025-S7359A" 2025; "6 CRR-NY 190.8 g" 2022). Regulations on protected plants in the state are more stringent with even more severe consequences ("6 CRR-NY 193.3 e" 2022). As such, even if one uses a decision support tool that directs a user to approved breeders, defects due to difficulty of breeding and propagation may be present in available native plant samples

(Kramer et al. 2019). There seems to be a need for connections to licensed, reputable vendors of native plants, as well as clear methods to preserve the biodiversity of threatened organisms.

2.3 Accessibility of Information and Resources

Effective communication of knowledge about restoration, biodiversity, and ecology is necessary for public integration and stewardship. Bridging the gap between academic publications and effective dissemination across various media is a difficult task. Connecting scientific work and data to common life takes effort that scientists not trained in public communication may struggle to accomplish (Geschke et al. 2023). Should statements lack clear language and come across as confrontational to readers, the author's intent could be clouded or ignored. Practice of science journalism regarding biodiversity can greatly ease the burden on scientists themselves (Geschke et al. 2023).

However, a failure to communicate information and intent may result in ecological illiteracy (Hooykaas et al. 2019). Disconnection from nature and a lack of field-specific education on biodiversity's importance can distance the public from the world around them (Hooykaas et al. 2019). Plant blindness, in which local and common plants are unidentifiable to residents of an area, is common in high-income countries, possibly due to a decline in exposure to and interaction with those plants (Stagg and Dillon 2022). With plants and animals in the natural environment culturally and physically removed from public access, knowledge withers and concern for that environment does as well (Hooykaas et al. 2019; Stagg and Dillon 2022). Local knowledge which was lost can benefit efforts to recreate interest in biodiversity, but should be informed by factual study (Joa et al. 2018). This includes indigenous and traditional knowledge bases, where it is easily polarizing to discuss the difference between established generations of practice and scientific experimentation and classifications (Joa et al. 2018). Approaching biodiversity restoration from an educational perspective including open integration with local communities may inspire better cooperation between scientists and non-scientists.

Information on biodiversity and native species restoration is accessible online, but clear communication is key to a successful outcome. General biodiversity products exist from the efforts of the scientific and internet communities to digitize observations (Saran et al. 2022). The development of these sites is far from complete, and listings may be limited by lack of descriptions and differing data standards (Saran et al. 2022). Inclusions of data from citizen science reports contribute to data disparity and is often highly dependent on an area's accessibility (Mandeville et al. 2022). Translating this information into spatial data is highly dependent on data quality and the architecture of the webpage in question (Mandeville et al. 2022; Rowland et al. 2020).

Additionally, web design is something that traditional geographic information systems (GIS) have more recently focused on. Applied technologies continue to shift towards the spread and availability of information through cloud data sharing and GIS web portals (Rowland et al. 2020). Increasing user familiarity with GIS services and therefore trust in their accuracy contributes to broadening of the applications of web mapping (Skarlatidou et al. 2013). Including transparency in GIS endeavors should be part of the process of creating tools for non-experts. This self-service GIS has several considerations to meet, as a wide user base must be able to access it. This includes those who are unable to process visual information, among other groups whose technical literacy is a limiting factor (Medina et al. 2015). Full development of web-based GIS portals would require greater attention and user testing from potential users. This would increase the interaction with such tools and put them to better use than if they remained sterile and technical.

3. Methods:

3.1 Review of Plant Presence Sites: Content and Construction

Several organizations have developed websites wherein listings of native plants are made, some with recommendations of preferred types for different reasons. To better understand how these sites organize and display their information, a review of present features was conducted. Users' capacity to directly compare features against other plants was desired for the project, as simply listing a name does not adequately describe a plant. Plant data for the current project was also needed, and therefore a comprehensive, easily retrieved, and responsive site was desired in order to contact and request permissions to use that data. Data organization, location data, and available traits also had to be assessed as the aim of this project differs from some sites. Several sites and guides were reviewed to compare approaches and inform the direction of this project.

Table 1. Plant database websites categorized by eight useful site characteristics.

Site Title & URL	Creator	Habitat Filters	Native Species Only	Map	Species Images	Trait Filters	Filter by County or Zip	Outside Links	Education Focus
Bplant Search	bplant.org	Regions Only	Yes	Regions Only	Yes, Not Complete (Progress Shown)	No	No	No	Yes
New York Flora Atlas	NYFA	No	Yes Specifies Naturalized	Static maps	Yes, Not Complete	No	Yes	Yes other eco sites	Yes
Occurrence Search	GBIF	No	No Presence Absence	Yes Live	Yes	No	Yes	Yes other eco sites	Yes
North American Keystone Plants	Homegrown National Park	Eco-region	Yes	Static maps	Yes	Yes	Yes	Yes other eco sites	Yes
Observations	iNaturalist	No	No, cannot filter	Yes	Yes	No	Yes	Yes other eco sites	Yes
BONAP	John T. Kartesz	No	Yes	Static/ Limited	No	No	No	Yes eco and gov	Yes
Native Plant Finder NWF	Justin Meissen NWF	No	Yes	Limited	Yes, Not Complete	Growth Types	Yes	Yes other eco sites	Yes
Special Collections: New York | NPIN	Ladybird Johnson Wildflower Center	Eco-region	Yes	No	Yes	Yes	No	Yes other eco sites	Yes

NatureServe Explorer PRO	NatureServe	Yes Biome & eco-region	Yes	Yes	No	No	Yes	No	Yes
Native Grass Seed	Prairie Moon Nursery	Region	Yes	No	Yes	Yes	No	Yes	Yes
NWPL-Species by Region NCNE	National Wetland Plant List USACE	Only Wetland	Yes	Static/ Limited	Yes, Not Complete	Yes	No	No	Yes

Most common were the webpages with listings of plants without direct map associations. Choosing an ecoregion or state was the defining factor which grouped these sites, yet some recommendations were not further divisible by area (Prairie Moon Nursery 2025; Ladybird Johnson Wildflower Center 2025; Homegrown National Park 2025). These sites also display imagery of chosen plant types, and allow the form (herb, shrub, etc.) and growth preferences (shade, moisture, soil, etc.) of plants to be filtered as well. There also is a focus on plant bloom time, which further cement that the main purpose of some sites is aesthetic choices of native plant types. The sites also are or have various links to vendors of native plants for the selected region. This seems to encourage purchase of native plants or simply ease to those who use the sites for that purpose.

A few native plant resources simply listed plants by species name for selected regions. While some sources were interactive (bplant.org 2025; Justin Meissen 2025), others were static documents (National Wildlife Federation 2025; Wild Ones of Western NY 2025; Cornell Cooperative Extension 2025). These listings seem to cater to people with an already established interest in wildlife and conservation, as direct imagery was not a feature. The interactive listing on Bplant.org had subpages that allowed exploration of imagery and range maps as well as growth habits of plants. There did not seem to be a way to filter other than by name. The Native Plant Finder under the National Wildlife Federation (NWF) is able to filter by zip code and plant type, but ranks these plants by the number of butterfly and moth species which used them as hosts for their caterpillars (Justin Meissen 2025). For the static documents, the NWF publication models after its interactive site, which is pollinator focused. The Cornell Cooperative Extension (CCE) publication also lists solely pollinator host plants. Wild Ones of Western New York (WOWNY) guide optionally lists pollinator types as well as local sources where the plants can be purchased, as well as whether it is a local ecotype or species.

Two plant search sites list occurrence data but seem not to allow filtering for specifically native species (GBIF 2025; iNaturalist 2025). Both the Global Biodiversity Information Facility (GBIF) and iNaturalist are focused on citizen science reports, with presence or absence capacity and subpages for plant listings. The option to determine if a plant was introduced is an option in the iNaturalist webpage, but it is unclear if such an option exists for the GBIF page.

The search tool for New York Flora Atlas (NYFA) was able to distinguish native plants and several other factors including naturalization (David Werier 2024). It was organized and digitized in conjunction with the New York Flora Association. The not-for-profit site also lists global ranking and endangerment status and allows splits into individual species pages. Individual pages also include distribution and the search itself is filterable, with search exclusions, by county. It accounts for accepted names and synonyms as well. There is a tool to compare records of species, allowing multiple entries to be visually compared by a distinct image of them, should the site host one. Individual plant pages cite continually updated herbarium records for the included plants, and link to various related websites (Table 1). Due to these traits and permissions granted by the organization, the information from NYFA was selected as the source for the plant data within this project.

Development of a sufficient tool which considers the preceding literature required flexibility best offered by coding a website. The language used is R, a popular statistical analysis language with a capacity to parse various datasets and spatial information. Many user libraries exist which can be called upon for additional functions which are not built into the base code. These libraries are available to be referenced in the code itself. Spatial analysis, mapping, and user interface modifications were needed to make the webpage usable. The code was finished in RStudio, a development program for R which allows visual display and versioning in conjunction with GitHub controls. GitHub allowed web hosting of the code and website, which was published as an R quarto markdown page.

3.2 Informal Interviews

Multiple parties were genuinely interested to hear of the possibility of such a tool when consulted. Formation of this idea has evolved over the past three years, with many people both in and outside of conservation consulted to choose the best features and scope for the project. This included an interview with representatives from the WOWNY chapter based in Amherst, New York, who participated in a consultation on the premise of the project and its functionalities (Joseph Han et al., "Wild Ones Western New York Consultation," May 22, 2025). This led to identification of various plant trait selections that were important in the project's construction. The consultation also identified a somewhat complex area for the project to be based on and narrowed the project from its initial county-wide scope.

3.3 Study Area

The regional focus of this project is the Town of Amherst, NY. Amherst is a suburb of 53.583mi^2 and 129,595 people ("GEOINFO: Area (Water, in ... in 2024 - Census Bureau Map" 2026). The Town stretches slightly longer North to South, and is bordered to the north by Tonawanda Creek, and the City of Buffalo and Town of Cheektowaga to the south (Figure 1). In the town, 20.22% of its land cover is wetland and hydrologic features ("Town of Amherst, NY

Mapping" 2025). There are two major and three tributary creeks, and the drainage of various sections of former wetland into developed suburban areas is colloquially documented throughout the town. Unique to the area is a silver maple-ash swamp named the Great Baehre Swamp Wildlife Management Area (Fig.1, green dashed area) that is designated as a Natural Heritage Community by the New York State Department of Environmental Conservation (NYSDEC) ("Natural Heritage Communities" 2025). It is classed as a vulnerable area in New York State and is known as a large area high quality occurrence of this ecosystem type ("Silver Maple-Ash Swamp Guide - New York Natural Heritage Program" 2025). Due to the high value of wetlands and hydrological connectivity, the project focuses on wetland proximity and native plant suitability for locations within Amherst, New York.

4. Data Sources

This project used soil data, wetland delineations, and species characteristics to develop recommendations for various environments in the study domain. The proximity to water, soil moisture retention, and a plant's preferences for these locations are important. Similarly, the growth habit of various plants, their annual or perennial status, for example, affects the permanence and location where these species should be located on a property. Perhaps keeping a low cover plant is more effective next to a home in the shade, and placing trees farther from the home can forestall limbs damaging the roof at some point. The utility of the chosen data in enumerating various characteristics is explored more below.

The NYFA native plant records were selected due to their completeness and their easily exported and thorough file keeping. Permissions were granted for this access, and a download option for the site's listings was available in a comma-separated value format in the advanced search selection. The site offered 1,082 records under the native plant category for Erie County, New York. These records were filtered to remove threatened, endangered, and rare species in Erie County. Due to New York State law, permissions are required to in any way transport or alter the growth of these species (“6 CRR-NY 193.3 e” 2022). Data cleaning reduced the records to roughly 973 species, and caveats were made regarding species composition based on their usability. It was not feasible within the time frame of the project to individually find and filter all organisms which may be irritable to or inedible by humans. A disclaimer of this is provided on the website for clarity.

Town boundary and hydrologic data was also retrieved from the Town of Amherst mapping portal (“Town of Amherst, NY Mapping” 2025). The products within are derived from other sources, such as the NYSDEC’s informational wetland areas listing, and the New York Natural Heritage Program’s Natural Heritage Communities map. Soil products are a derivative of Soil Survey Geographic Database (SSURGO). Soil consistency and wetland adjacency through permeation are both listed characteristics of the dataset. Creating geojson files from layer files in ArcGIS took place through exporting, then the resulting tables were filtered to usable fields in R.

5. Data Processing for Site Implementation

Raw data is unsuitable for display as unassociated site data does not usefully reflect how that information can be used to direct user selection based on habitat suitability. This project is constructed through ArcGIS, R, and the dependent libraries and tools of each with a focus on replicable research structure. The initial wetland proximity analysis was completed in ArcGIS Pro and is detailed below. As this project’s restoration focus is the selection of wetland suitable plants, a weighted map of wetland proximity for critical connectivity areas was created. First, the shapefiles for hydrology, drainage ditches, State Informational Wetlands, and the Great Baehre Swamp were added to a map file. Then, multi-ring buffers were added around State

Informational Wetland areas at distances of 150, 300, and 600 feet based on a wildlife buffer study, prioritizing proximity to wetland habitats (McElfish et al. 2008). A visual-only buffer of ½ mile was created for illustration purposes around the Great Baehre Swamp (Figure 1). The Hydrologic Features layer received the same distances but smaller weights due in part to the focus on wetland plant connectivity. The central layers of State Informational Wetlands and Hydrologic Features were given weight based on the base 10 logarithm of their area divided by 2 and rounded to the nearest whole number. This was done to normalize the areas of each parcel such that their assigned weight was comparable, while still ranking them based on overall area. Small features were given a value of 1, including ditches along roads which collect runoff and are thus able to host plants of some wetland tolerance. This considered larger wetland features as more critical to wetland protection due to their larger hosting capacity.

Each buffer ring was assigned a weight according to distance and relation to wetlands. The State Informational Wetland buffers were assigned to a value of 4, 3, and 2 from inner to outermost. Similarly, the Hydrology buffers were given values of 3, 2, and 1, as wetland plant connectivity specifically was the focus of the score. The wetland weight was all appended in the same fieldname of the various attribute tables. Then, after translating these vector shapefiles into a raster, each field summed to a final raster product (1750 x 2475 cells, 30 x 30ft. per cell) of wetland proximity value. This value was then correlated to rough categories of plant wetland occurrence as outlined by the National Wetland Plant List in the table below (Minkin et al. 2023).

Table 2: Definitions Retrieved from 2022 National Wetland Plant List U.S. Army Corps of Engineers (Minkin et al. 2023)

Score	Wetland Indicator Status Rating	Definition
0-2	Upland (UPL)	Almost never occur in wetlands
3-4	Facultative Upland (FACW)	Usually occur in non-wetlands, but may occur in wetlands
5-6	Facultative (FAC)	Occur both in wetlands and non-wetlands
7-8	Facultative Wetland (FACU)	Usually occur in wetlands, but may occur in non-wetlands

9-10	Obligate (OBL)	Almost always occur in wetlands

The scores of 0-2 were associated with Upland, 3-4 with Facultative Upland, 5-6 with Facultative, 7-8 with Facultative Wetland, and 9-10 with Obligate Wetland. Scoring wetland proximity as similar to wetland abundance is due to conditions needed to gauge plants as various wetland grades and the spatial autocorrelation in physiochemical conditions near wetland areas (Churko et al. 2020; Kraft et al. 2019; Lichvar 2012). This scoring is used to connect plant types via their indicator status code to the proximity value generated by the map. The value generated is an attempt to combine wetland proximity through distance and increasing connectivity. Placing larger bodies of wetland features at a higher rank is partly due to higher hosting capacity, but also in hopes that larger areas can facilitate more connectivity and therefore ecosystem stability. Being closer to an area allows more interactions with that area; illustrating that to a userbase is important to show how the environment around them interacts with their choice of plants.

The proximity value raster was initially exported in tiff format, and the rest of the wetland-adjacent files were in geojson form. The buffer around the swamp and the boundary for the town of Amherst and the SSURGO data remained in vector format. It was later decided that while the tiff format cleanly showed wetland proximity value, it was clearer to use a hover tooltip function from a vector union of the layers to display single values above locations more accurately. The hydric status of the soil and soil name are continued over from the SSURGO data, identifying the potential for wetland inclusions in each soil section ("NRCS State Hydric Soils List" 2026). I used RStudio to transform the map layers to the correct coordinate projection, WGS84. The Leaflet library was used for the interactive map and layer legend (Cheng et al. 2025).

Invalid data and doubled entries for native plants were fixed by filtering and cleaning the dataset. The filtering involved dividing the dataset into separate classes due to NYS regulations for rare, threatened, endangered, and extirpated plants. The remaining list of 973 common plants was then cleaned through column name editing and the separation of species which had multiple traits by growth habit. Several classifications for a single species would break the table and search interface. This split the entries into 1052 observations, with some repeats due to growth class. Then, the interactive table with filtering functionality was created using the Crosstalk library (Cheng and Sievert 2025). Several plant traits from NYFA were used to filter species, including coefficient of conservatism (later renamed as difficulty), growth type, growth permanence, wetland status, common name, and scientific name. Renaming coefficient of conservatism to difficulty was completed to relate how specific the environmental tolerances for each plant is required for successful establishment, directly relating to the effort and difficulty of matching the plants to those preferred conditions. There is also an about page and a functionality page, which go through the motivations and coding process for ease of replication by others (Figure 1-3). The finished product is hosted on GitHub and is not resource-intensive, acting as proof of concept that it is possible to create this type of resource with publicly available data and methods.

Replication of the project was a major consideration in development. The goal was to create a rough scaffold which could exchange data from different areas with relative ease. In the spirit of that effort, most steps taken in this project are documented, whether in the GitHub hosted R code, or here in the accompanying paper report.

6. Results

The website hosting the web-tools is published to GitHub Pages through Quarto of the open-source Public Benefit Corporation Posit, and was coded through the development environment of RStudio (Posit team 2025). The page has a navigation bar with three locations: ‘Home,’ ‘How it

Works,' and 'About.' These refer to the main page, a sub-page which describes how the site is coded and what sources are used, and an about page with a brief description of the development conditions and goals of the site. On the home page above the tools themselves there is a quick description of how to utilize the webtools included through navigation of the map, identifying what conditions are displayed, and then browsing the plant suggestions in the table below it.

The map tool uses leaflet to call the layers which are included and may be toggled through a collapsible menu at the top right corner of the map (Cheng et al. 2025, Figure 3). The layers include an outline of the Town of Amherst's boundary, a Significant Wetland buffer layer, an interactable Soil and Wetland Value(SWV) shapefile layer, a deselected raster layer, and a toggle-able legend of the displayed Wetland Values. Upon hover with the cursor, the collapsible menu unfolds and allows selection of each layer through a radio toggle button. Hovering the cursor over the SWV layer creates a hover tooltip which reads "This section has a Wetland Value of X. Soil Y 'SOIL NAME.'" Within, the x value is the Wetland Value for the area under the cursor, the Y value is the hydric status of the soil as defined in SSURGO, and the 'SOIL NAME' value is the dominant soil series of that area. The Wetland Values legend has a color scale and corresponds directly to the colors on the SWV layer.

Below this map tool is a table of wetland codes and classes, essentially the structure and content of Table 2 in this paper. Users are to take note of this section and use the estimated wetland value to find what wetland status they should explore in the plant table below. A small reminder that plants may not be suited for every case exists above the table as well. The table itself is organized by six columns: Scientific Name, Common Name, Difficulty, Wetland Status, Growth Habit, and Duration. The filters for Wetland Status and Duration are selectable from a dropdown list and can be deleted by pressing the backspace key. Growth Habit is organized as a series of radio buttons in two columns, and multiple types can be selected at once. Difficulty is actually coefficient of conservatism, which corresponds to how wide a range of ecological tolerances the plant can withstand (*Floristic Quality Assessment Index* 2014). This can be filtered through a slider from 0 to 10, with 10 being most difficult to match ideal growing conditions. Duration describes how the plant grows, with perennial, biennial, and annual descriptions. The table may also be searched through the search bar above it.

7. Discussion

In this project I developed a decision support tool to help users find suitable native plant species for their specific environment within the town of Amherst, NY. Ultimately, the website meets several of the conditions of its intended development. The functionality of displaying wetlands in relation to locations in the Town of Amherst is realized. The tool uses soil type, hydrologic proximity, and wetland locations for a mapping system. Additionally, a plant selection system filterable by growth type, plant duration, difficulty of care, and wetland status was included to complement the map's display output.

Creating a tool where users can reference a map and choose plants based upon suitability to their environment required a longer web development process than anticipated. Choosing to forgo aesthetic selection through images of every native plant was mainly for the sake of storage space and development constraints. A large cache of high-resolution images would be time intensive to source and standardize, making it prohibitive to add into an early working version of the project. Finding accurate free-source photos of each plant type would require time and resources that were not available for this tool's development. Using GitHub to host means that size caps on build and the amount of traffic to and from the webpage are limited, which means it would not be suitable for a large userbase. Hosting requirements on GitHub also affected the coding process in what complexity of code was feasible to host, as something running on R's

Shiny library dependencies would be intensive and therefore likely prohibitive for many to access.

Access to the website was at one point considered to have a user input function and therefore an account and verification process. This was discarded as impractical to implement, and the aim was accessibility to information rather than users changing the site itself. The eventual concept of the user matching their desired location with the correct wetland intensity and preferred plant type was much farther in the process than initially planned. After careful study of examples such as iNaturalist and GBIF it seemed prohibitive to make a web repository of user-created observation points for local native plants.

Organizing data such that facultative wetland codes are the main source of recommendations may require further work. Ideally including direct water needs and soil requirements by each plant type and linking those to the corresponding fields would greatly improve functionality. The search functions that filter the string data can now sort some of the data, yet with how incomplete some of the other data fields were for plants in the full set, those fields had to be removed.

More user experience and interaction with the webpage would be helpful in assessing the success of this tool. A testing phase with constructive feedback and user accessibility would be very helpful. Unfortunately, due to time and resource constraints in this project, this was not possible. It would also be prudent to re-analyze the site's accessibility features to get a more complete picture of how it works for people who need them. Alterations of naming, titles, and display types could be under consideration.

It also quickly became clear that the project could not become a true scaffold for other areas. The site's structure is very dependent on the area of study, and proliferation of it requires further streamlining. An ideal scaffold design would allow input of data given a consistent format, which at this time seems prohibitive due to a lack of standardized species information on a locational basis. Creating a section of code to import and restructure native plant species data would be a project of itself given the diverse methods of record-keeping in botany. Communication between environmental institutions on a far grander scale than the current scope of this project would be ideal.

Hopefully, the clarity and the relatively uncluttered nature of the final product provide a simple user experience. The creation of this tool is done in the hope that someone may find use of it, and that others may use its structure for their own similar endeavors. Localized native planting efforts can be better coordinated with adequate information and resources. Increasing plant literacy and interconnectivity of wetland ecosystems are both extremely desirable outcomes of this webpage and appended report. If the intent of this project holds true, it will be used to these ends.

8. Maps & Figures

Figure 1: A map of State Informational Wetlands and a State Significant Wetland feature buffer

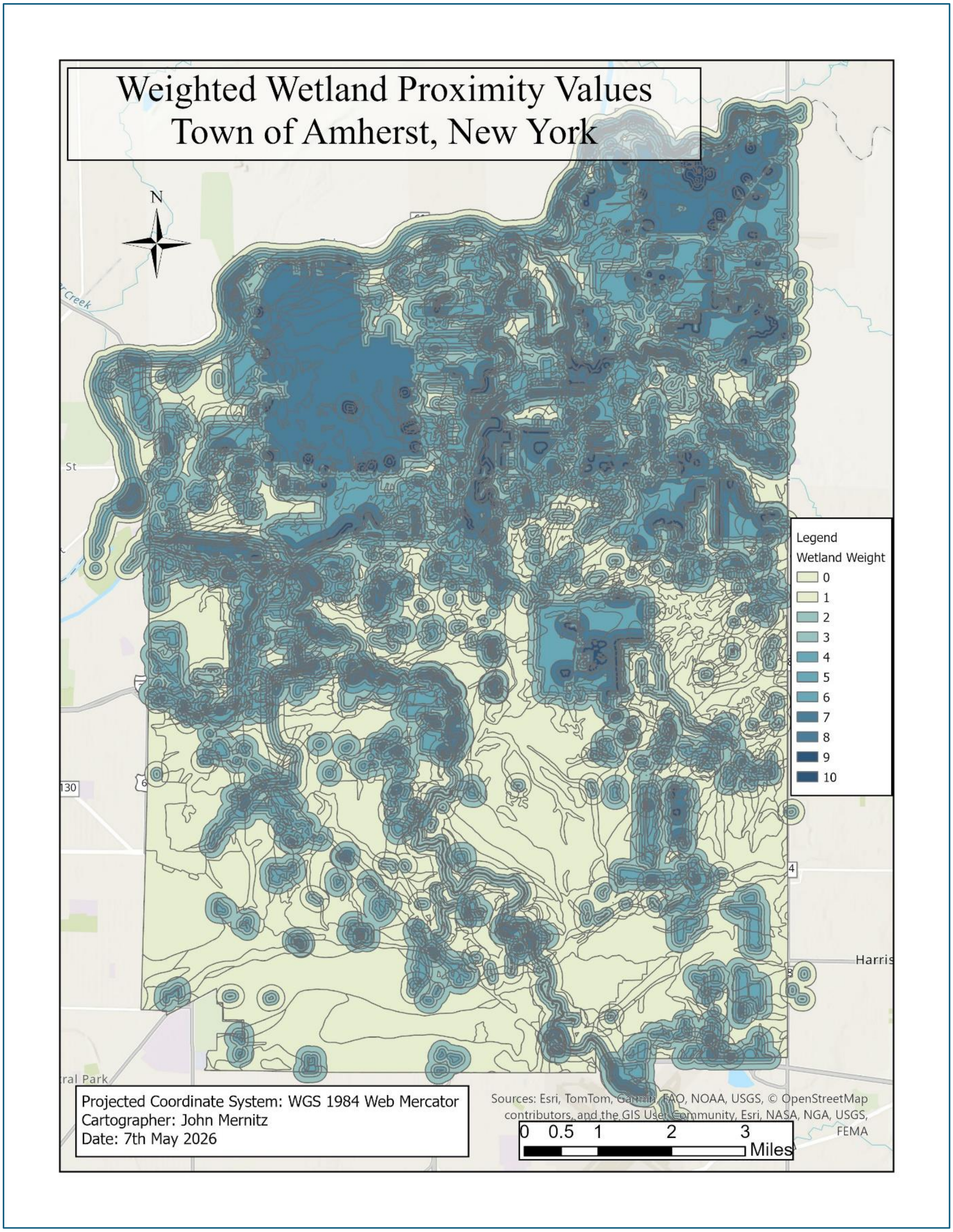

Figure 2: A vector layer map of State Informational Wetlands, SSURGO soils, and Hydrologic features. The displayed value is given as Wetland Weight.

Figure 3: A Screenshot of the Map on the Home Page of the Site

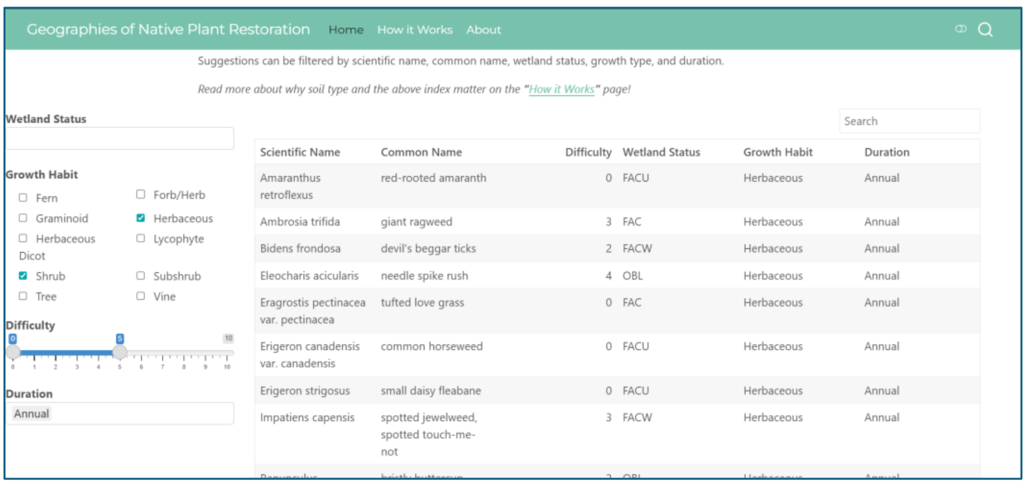

Figure 4: A Screenshot of the Search Filters and Table on the Home Page of the Site

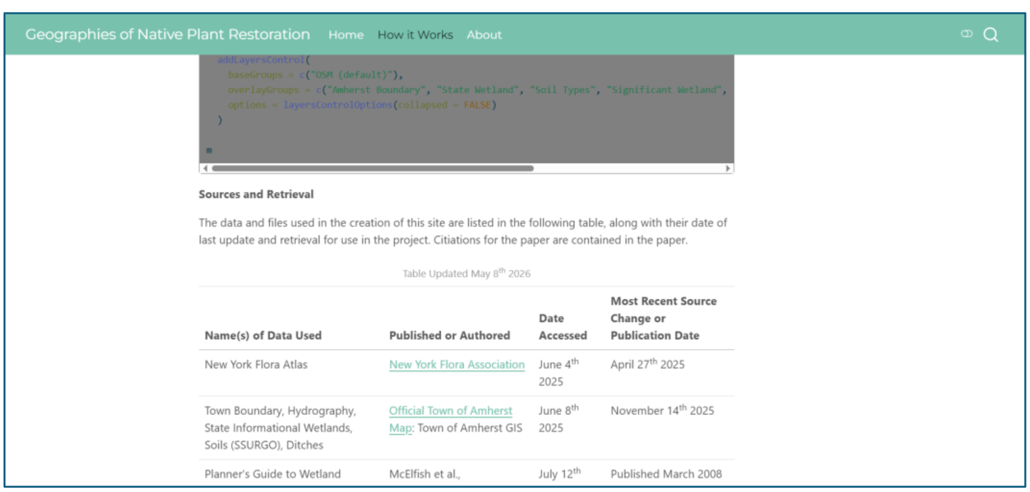

Figure 5: A Screenshot of Some Code and Credits on the How it Works Page

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